

OpenClaw Temperature and Humidity Dashboard Tutorial

Before starting, please complete the following from the `Pre-use Preparation` directory: [01-Peripheral Hardware Configuration], [02-OpenClaw Jetson-KEY Configuration], [03-OpenClaw Model Switching], [04-AI Voice Interaction Configuration], [05-Three Dialogue Solutions Configuration Test]. If you are currently only using TUI / WhatsApp, you can skip `04`. This article assumes the basic environment is already prepared and only covers the Temperature and Humidity Dashboard project itself.

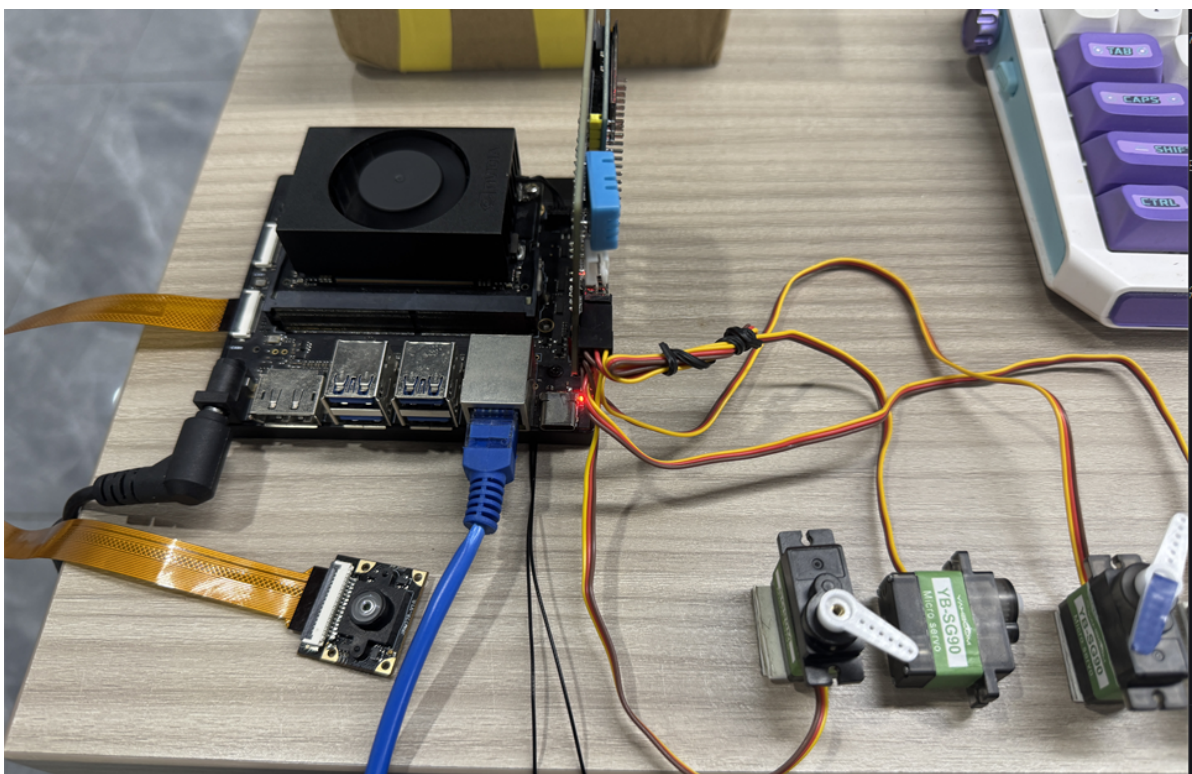
1. Tutorial Objectives

This article demonstrates OpenClaw's multi-peripheral coordination capabilities in a desktop dashboard scenario through the combination of "sensor + display + indicator". After completing this tutorial, readers will be able to:

- Understand how to combine DHT, OLED, RGB, and servo into a temperature and humidity dashboard
- Interact with OpenClaw through three methods: WhatsApp, TUI, and voice code
- Complete four project scenarios: real-time readings, OLED indicator panel, RGB status light, and servo pointer mapping

2. Hardware Preparation

- OpenClaw main control board
- DHT temperature and humidity sensor
- OLED display
- RGB light strip / RGB LED module
- [Optional] AI voice module + speaker
- Wiring diagram



3. Software Preparation

Code entry points used in this tutorial

- Temperature and humidity reading command: `/home/jetson/hardware_hal_demo/dht`
- OLED control command: `/home/jetson/hardware_hal_demo/oled`
- RGB control command: `/home/jetson/hardware_hal_demo/rgb`
- Servo control command: `/home/jetson/hardware_hal_demo/cli/servo`
- Voice entry: `/home/jetson/voice_to_openclaw/src/main.py`

This project is very similar to the AI Weather Station, but with a different focus:

- The AI Weather Station emphasizes "broadcasting" and "interpretation"
- The Temperature and Humidity Dashboard emphasizes "display" and "status mapping"

4. Testing Project Basic Capabilities

Before starting this project, please complete a basic `openclaw tui` dialogue self-check; if you haven't done so yet, see [05-Three Dialogue Solutions Configuration Test] first. After passing, proceed to the Temperature and Humidity Dashboard specific test.

It is recommended to first enter OpenClaw TUI:

```
openclaw tui
```

```
openclaw tui - ws://127.0.0.1:18789 - agent main - session main  
  
session agent:main:main  
connected | idle  
agent main | session main (openclaw-tui) | baillan/qwen3.5-plus | tokens 0/1.0m (0%)
```

Then test the following types of commands in sequence:

- Read current temperature and humidity
- Display current temperature and humidity on OLED
- If the environment is normal, turn on green light
- Turn the servo to 90 degrees

5. Three Solutions for Communicating with OpenClaw

Solution	Suitable Scenarios	Advantages	Recommended Use
WhatsApp	Remote status viewing, classroom demonstrations	Convenient for remotely reading current panel data	Demonstrate "remote monitoring panel"
TUI	Local debugging, mapping verification	Easiest for troubleshooting data flow between sensors and actuators	Run through the visualization pipeline first
Voice code	Voice broadcast, desktop assistant	Closer to daily usage methods	Use as the complete demonstration solution

If the optional AI voice module + speaker is selected, Feishu dialogue, TUI, and WhatsApp dialogue need to read aloud OpenClaw's reply content. You can run the following program in the terminal. The AI voice module dialogue does not need to run this program repeatedly.

- Enable the broadcast parameter

```
/home/jetson/voice_to_openclaw/.env
```

```
ENABLE_TTS=true #This parameter controls whether audio readback is enabled.
After modification, save and run the following program in the terminal
```

- Run the following in the terminal:

```
cd ~/voice_to_openclaw
source .env/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) jetson@yahboom:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/jetson/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
[播报] 开始播报 tui 回复
```

5.1 WhatsApp Solution

For WhatsApp integration and bot creation steps, please refer directly to [05-Three Dialogue Solutions Configuration Test]. After creation, you can remotely view dashboard status in WhatsApp, for example:

- Temperature and humidity dashboard

20:36



< 2



+86 153 3885 7526 (你)

给自己发消息

warm or if temperature continues to rise

Would you like me to set up a reminder if the temperature exceeds 30°C?

Note: I did not schedule a reminder in this turn, so this will not trigger automatically.

20:34 ✓✓

Temperature and humidity dashboard

20:35 ✓✓

Temperature and humidity dashboard is now active!



Current Readings:

- **Temperature:** 29.7°C (85.5°F)
- **Humidity:** 54%



OLED Display: Showing temp and humidity values

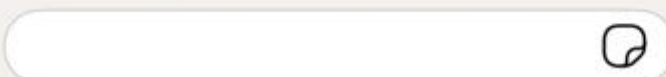


RGB Indicator: Red blinking light (speed 5)

- This indicates the temperature is **above 28°C** (warm/hot zone)

Status: The room is quite warm. Consider turning on a fan or AC if you're feeling uncomfortable!

20:35 ✓✓



5.2 TUI Solution

If you have completed the TUI self-check in article 0, directly enter terminal text dialogue:

```
openclaw tui
```

Then enter project-related commands, for example:

- Temperature and humidity dashboard

```
File Edit Tabs Help
The weather card is now showing on your OLED screen! ✓

、 Temperature and humidity dashboard

Temperature dashboard activated! 🌡️

Current Readings:
- Temperature: 30.0°C (86°F)
- Humidity: 52%

Actions:
- ✓ OLED: Displaying "Temp: 30.0C, Humidity: 52%"
- ✓ RGB: Red blinking effect (speed 5) - indicates high temperature (>28°C)

Status: ⚠️ Temperature is above the comfortable range (28°C threshold). The
red blinking light serves as a heat warning!

The dashboard is now running and will continue to show the current
environmental conditions. 📺
connected | idle
agent main | session main (openclaw-tui) | bailian/qwen3.5-plus | tokens
82k/1.0m (8%)
```

5.3 Voice Code Solution

Requires optional AI voice interaction module + speaker

For voice module installation, environment configuration, and basic wake-up procedure, please refer to [04-AI Voice Interaction Configuration]. After wake-up, you can directly speak control commands, for example:

- Temperature and humidity dashboard


```

[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 9990 ms 语音, 正在转写

[你] Temperature and humidity dashboard.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] ## 🌡️🌡️Temperature & Humidity Dashboard Active!

**Current
  Readings:**
- **Temperature:** 30.1°C (86.2°F)
- **Humidity:** 51%
  **Dashboard Status:**
- ✔️ OLED displaying real-time values
- ✔️ RGB
  indicator: **Red blinking** (heat warning - above 28°C)
  **Temperature Color Code:**
- 🌡️ Blue breathing: < 20°C (Cold)
- 🌡️
  Solid green: 20-28°C (Comfortable)
- 🌡️ Red blinking: > 28°C (Hot) ⚠️
  **Current Status:** **HOT** - Temperature is above the comfort
  threshold! Consider cooling measures. 🌡️

The dashboard will continue
  showing live readings. Want me to check again in a few minutes?

```

6. Project Implementation

Note: The following demonstrations are all conducted using `openclaw_tui` dialogue. You can also choose WhatsApp or voice methods to complete them.

6.1 Scenario 1: Real-time Readings

Experiment Objective

Read the current temperature and humidity to complete the basic data input for the dashboard.

Effect Description

OpenClaw will first retrieve the temperature and humidity readings, which serve as the basis for subsequent OLED, RGB, and servo mapping.

Example Commands

- Read current temperature and humidity
- What's the current temperature and humidity
- Help me refresh the current readings

6.2 Scenario 2: OLED Indicator Panel

Experiment Objective

Display the current temperature and humidity results as concise OLED dashboard content.

Effect Description

This is suitable for using `metrics` or `dht_oled` methods to combine `Temp` and `Hum` into a compact screen area, creating a small data panel.

Example Commands

- Display Temp and Hum on OLED
- Create a simple temperature and humidity panel
- Write the current temperature and humidity to the screen

6.3 Scenario 3: RGB Status Light

Experiment Objective

Use RGB colors to express whether the current environment is normal, hot, cold, or dry.

Effect Description

This step can make the dashboard more intuitive. For example, green for normal, yellow for hot, red for abnormal, blue for cold — you can see the status at a glance.

Example Commands

- If the environment is normal, turn on green light
- If the temperature is high, turn on yellow light
- If the humidity is abnormal, turn on red light

6.4 Scenario 4: Servo Pointer Mapping

Experiment Objective

Map temperature or humidity to servo angle, making the servo point to a position like a gauge pointer.

Effect Description

For example, map the temperature range 0 - 50 to 0 - 180 degrees. The higher the current temperature, the farther right the servo pointer moves. This effect is very suitable for classroom demonstrations, because "turning numbers into physical actions" is very intuitive.

Example Commands

- Use the servo to represent current temperature like a pointer
- Map the current temperature to servo angle
- Make the pointer point to the current status